



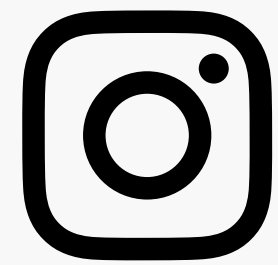
A HYBRID MARKOV CHAIN AND SHAPLEY VALUE APPROACH TO MULTI- TOUCH ATTRIBUTION AND BUDGET OPTIMIZATION IN DIGITAL MARKETING

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THE PROBLEM

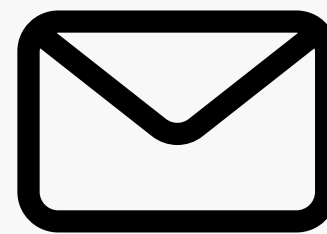
Who deserves the credit?



Instagram



Google Search



E-mail



Direct



Purchase

- Last-click attribution gives **100% credit to the final touchpoint.**
- All earlier channels: **zero credit**



THE REAL CHALLENGE

The Validation Gap

Many attribution methods exist.
But how do we know which one is **correct**?

- ✗ Real-world data has **no ground truth**.
- ✗ We can't ask the customer's mind: "what really convinced you?"

→ Attribution research has been a **decades-old debate** without a referee.



OUR APPROACH: SYNTHETIC GROUND TRUTH

Building a controlled laboratory

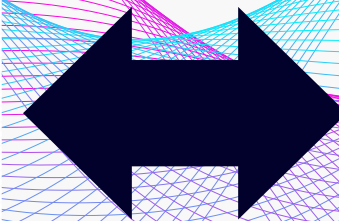
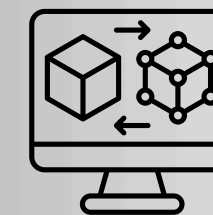
Synthetic Dataset

- 30,000 customer journey
- 5 marketing channels
- Calibrated from industry benchmarks
- 1.18% conversion rate



Known Ground Truth

- Channel effects pre-defined
- Monte Carlo (100k paths)
- All 8 models tested against it



→ Every method gets a **fair, objective grade**

THE HYBRID MODEL

Combining two complementary methods

Markov Chain

- ✓ Ranks channels correctly
- ✗ Exaggerates magnitudes

$$\text{Hybrid} = \alpha \cdot \text{Markov} + (1-\alpha) \cdot \text{Shapley}$$

Empirically optimal: $\alpha = 0.25^*$
(25% Markov + 75% Shapley)

Shapley Value

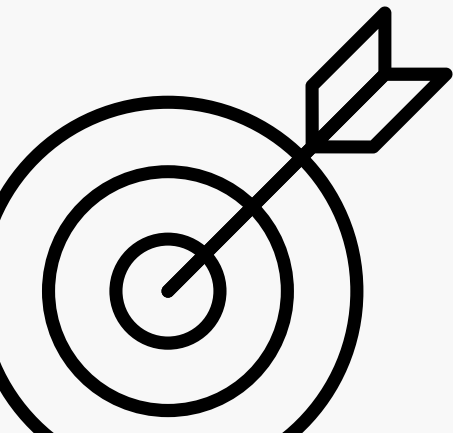
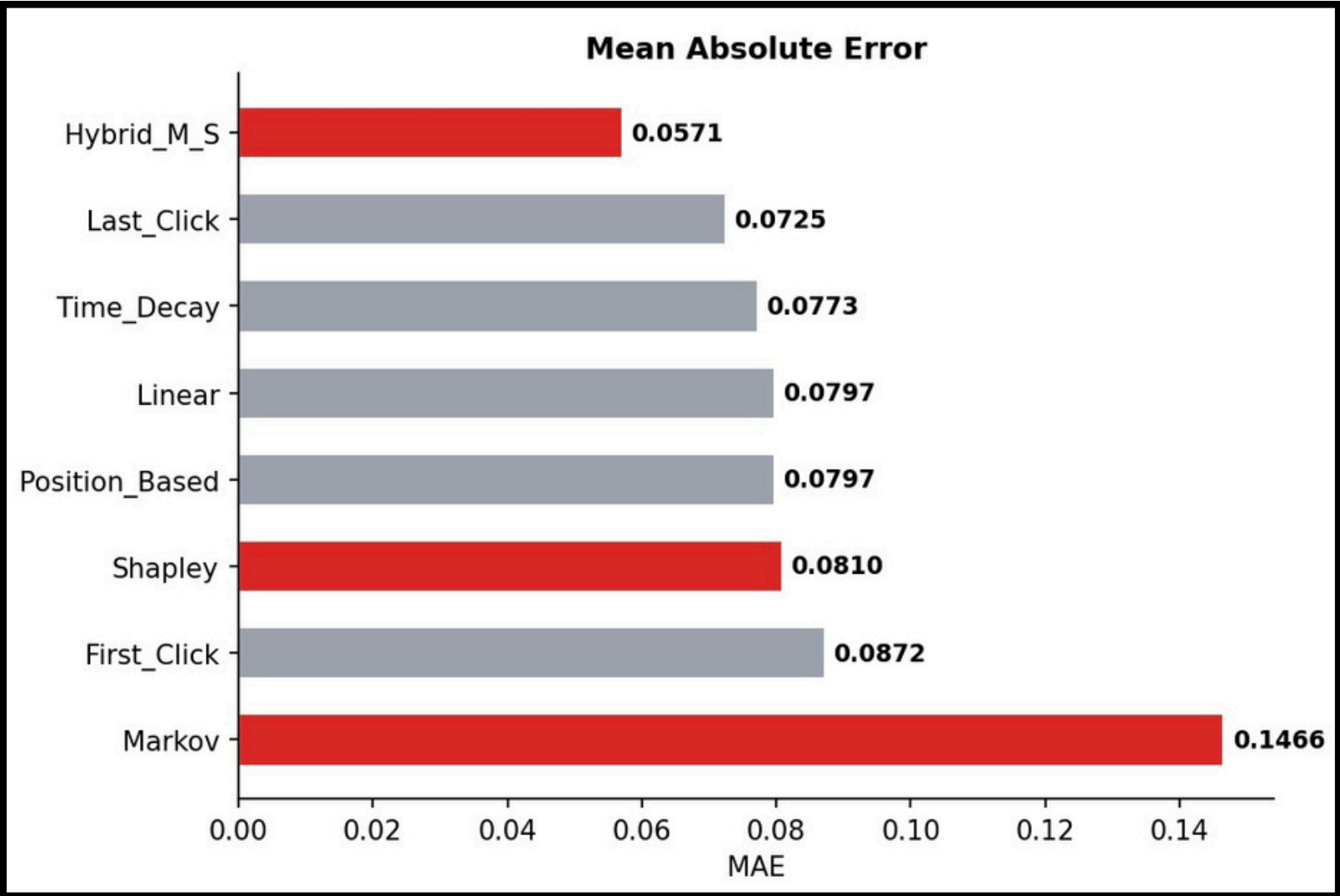
- ✓ Calibrated magnitudes
- ✗ Blurs the ranking

RESULTS: ATTRIBUTION ACCURACY

The hybrid model wins - by a clear margin

METRIC	LAST-CLICK	HYBRID
MAE	0.0725	0.0571
SPEARMAN P	0.300	0.600
RANK ACCURACY	40%	60%

→ 21.2% lower error, 2x better ranking

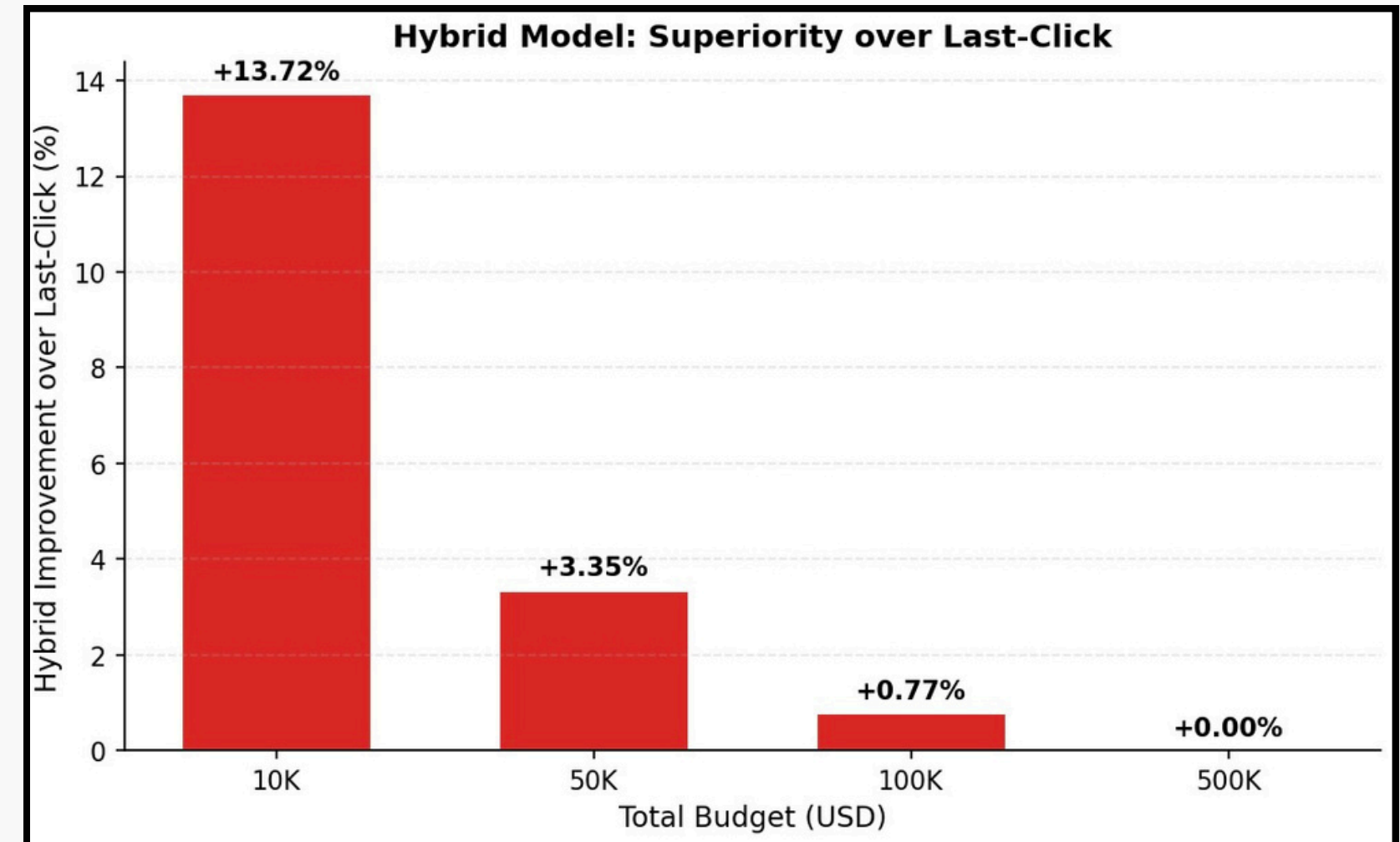


RESULTS: REVENUE IMPACT

What does this mean in dollars?

BUDGET	IMPROVEMENT
\$10,000	+13.72%
\$50,000	+3.35%
\$100,000	+0.77%
\$500,000	≈0% (saturation)

→ **Greatest value: small-to-medium B2C operations**



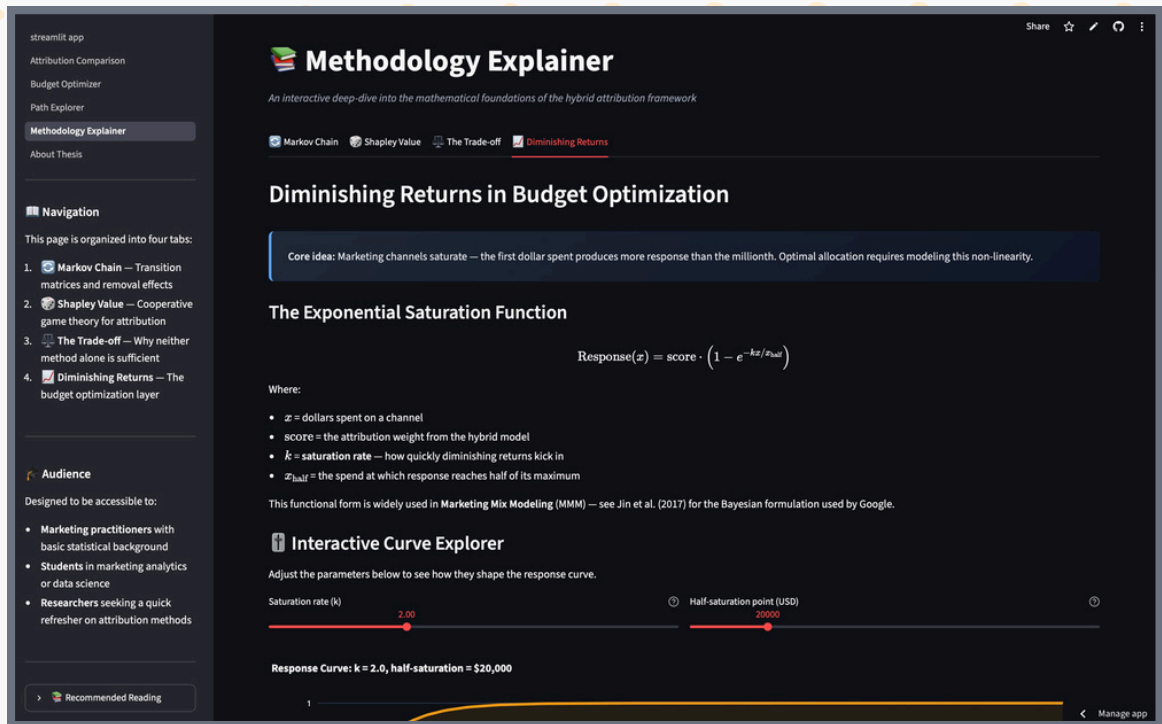
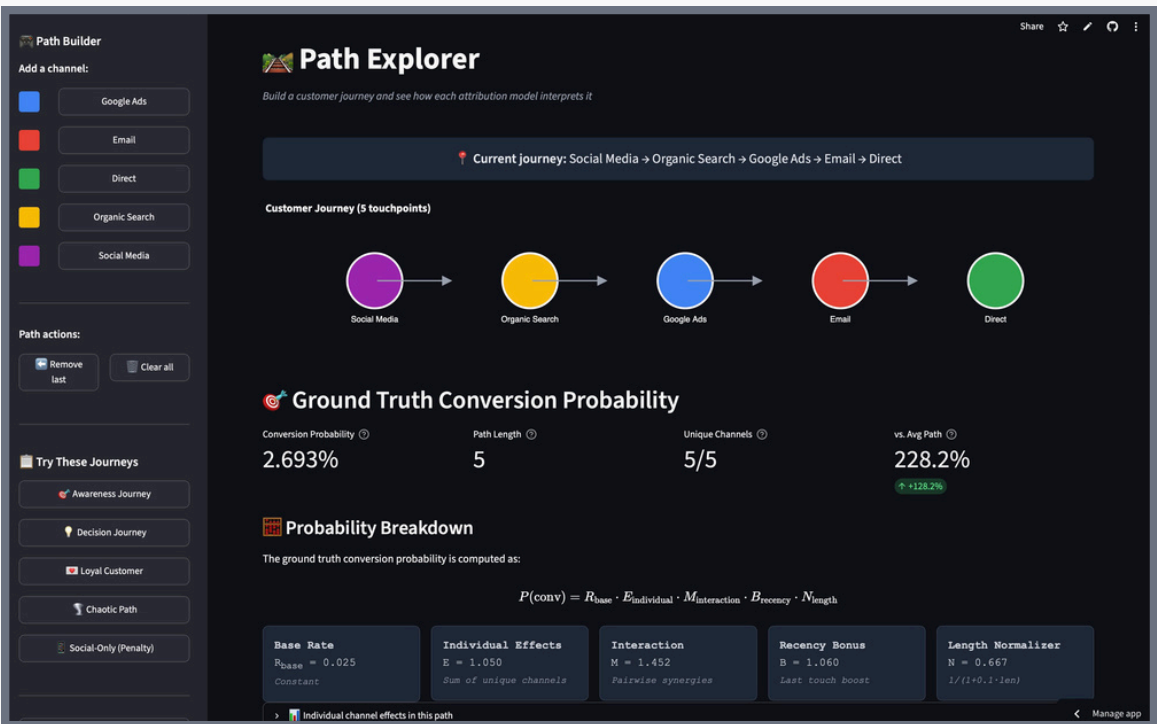
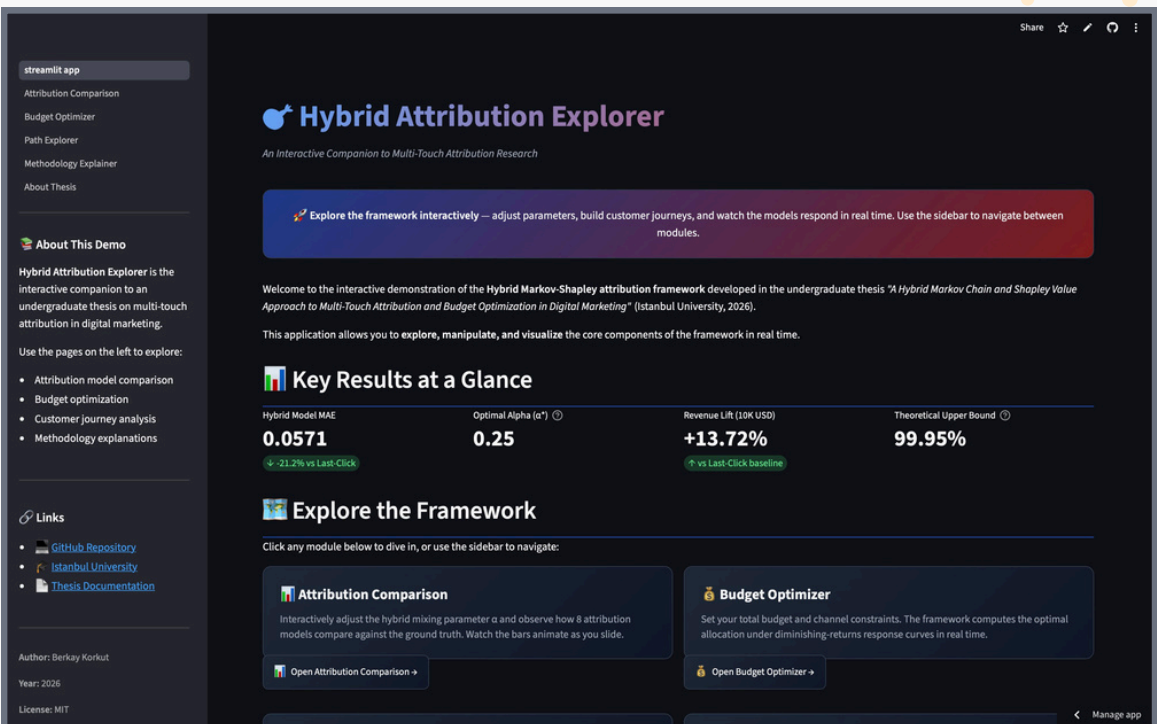
INTERACTIVE DEMO

An interactive companion -
explore it yourself

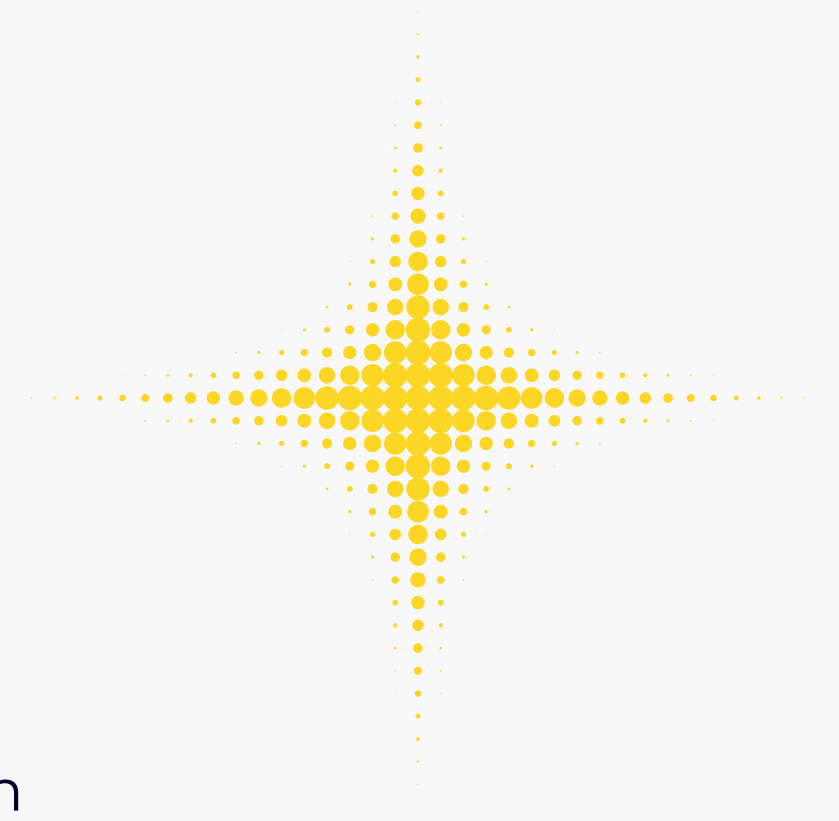


hybrid-attribution-explorer.streamlit.app

[GitHub: brky48/hybrid-mta-attribution](https://github.com/brky48/hybrid-mta-attribution)



THANK YOU SO MUCH!



Three Contributions

Validation framework – synthetic ground truth for objective comparison

Markov-Shapley trade-off – formally identified and resolved

Integration with budget optimization – attribution meets allocation

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